

A filter is a circuit that is designed to pass signals with desired frequencies and reject or attenuate others.

Filters

□ Passive filter: Passive filter consists of only passive element R , L and C .

- Filters are the circuits used in radio and TV receivers to allow us to select one desired signal out of a multitude of broadcast signals in the environment.

Applications: Radio, Audio electronics, Mobile Phone, TV, Power supplies etc.

Classification:

1. low pass filter: A low pass filter passes low frequencies and stops high frequencies, as shown in ideally,

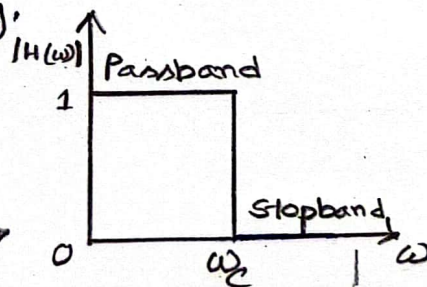


Fig. 1: Ideal characteristics of low pass filter.

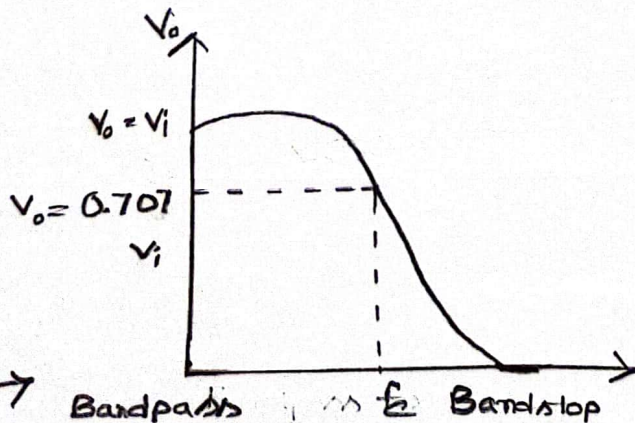


Fig. 2: Practical characteristics of low pass filter.

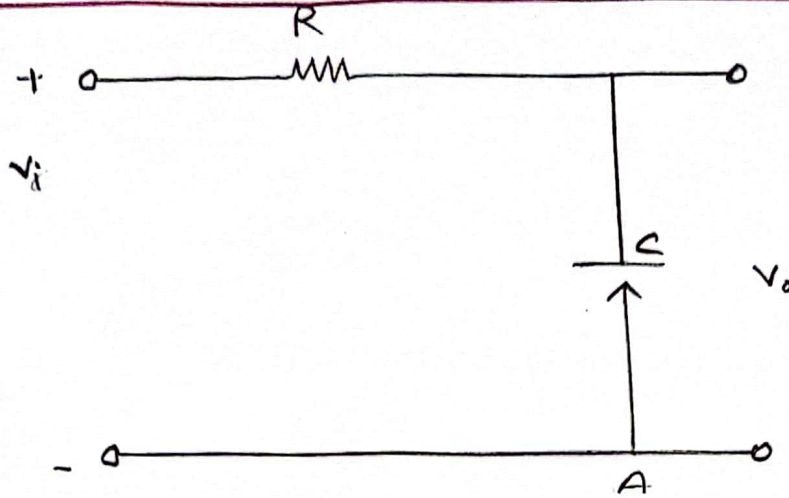


Fig-03. R-C low pass filter.

Now, At $f = 0\text{Hz}$,

$$X_C = \frac{1}{2\pi f C} = \frac{1}{2\pi \times 0 \times C} = \infty \Omega$$

Reactance

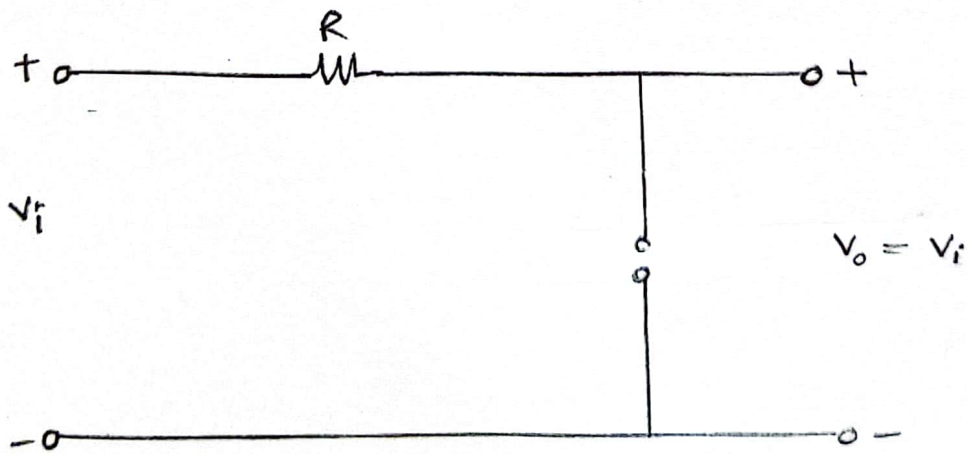


Fig-04.

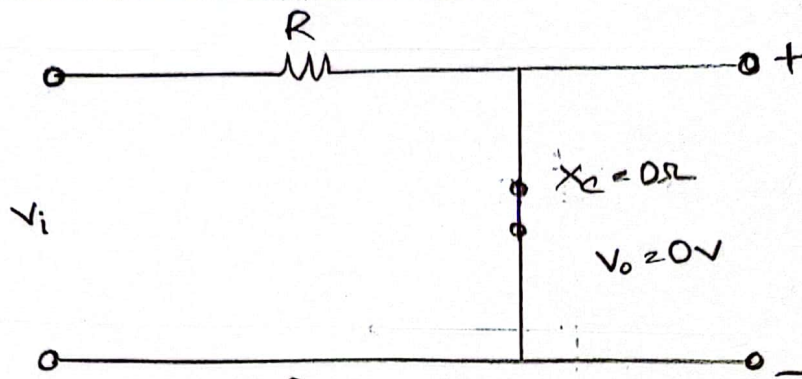


Fig-05.

□ Cut off frequency, $f_c = \frac{1}{2\pi RC} V$

□ Voltage gain, $A_V = \frac{V_o}{V_i} = \frac{X_c}{\sqrt{R^2 + X_c^2}}$

∴ phase, $\theta = -\tan^{-1} \frac{R}{X_c}$

2. High Pass filter: A high pass filter passes high frequencies and reject or attenuate others.

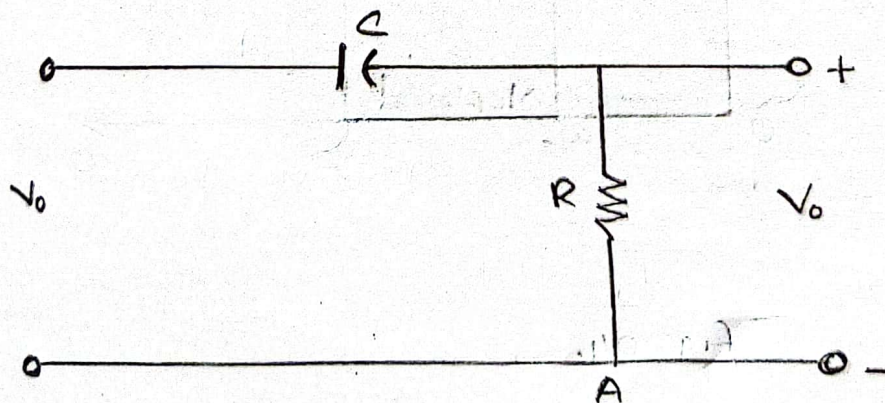


Fig-01. R-C High pass filter

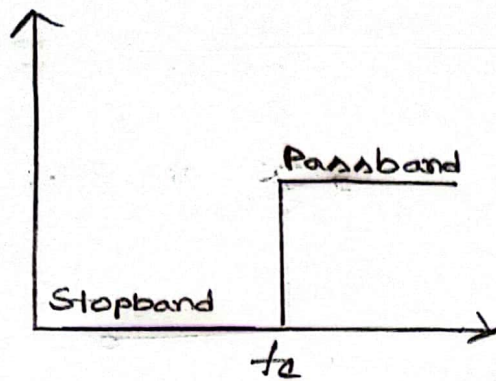


fig:02. Ideal characteristics of High Pass filter

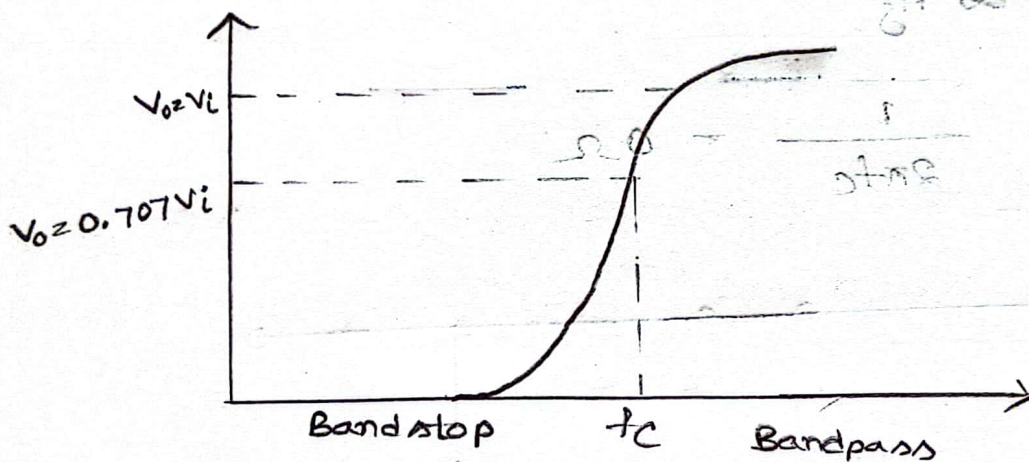


fig:03. Practical characteristics of High Pass filter

□ Cut off frequency, $f_c = \frac{1}{2\pi RC}$ V

□ Voltage gain, $A_v = \frac{V_o}{V_i} = \frac{R}{\sqrt{R^2 + X_c^2}}$

• Phase, $\theta = -\tan^{-1} \frac{X_c}{R}$

Now At $f = 0$ Hz,

$X_c = \frac{1}{2\pi f C} = \frac{1}{2\pi \times 0 \times C} = \infty \Omega$

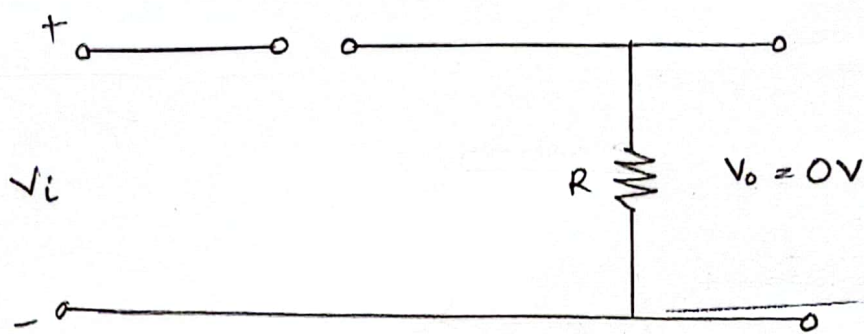


Fig-04.

At $f = \infty$ Hz,

$$X_C = \frac{1}{2\pi fC} = 0\Omega$$

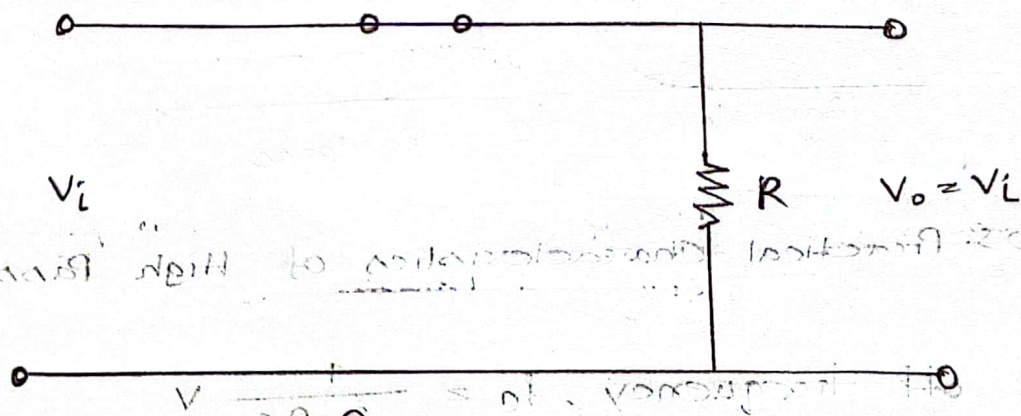


Fig-05.

Q. * Example: 22.61 Given, $R = 20K\Omega$ and $C = 1200pF$:

a. Sketch the normalized po plot if the filter is used as both a high-pass and a low-pass filter.

b. Sketch the phase plot for both filters in Part (a)

c. Determine the magnitude and phase of $A_V = \frac{V_o}{V_i}$

at $f = \frac{1}{2} f_c$ - low high pass filter.

Ans : a

$$\text{cutoff frequency, } f_c = \frac{1}{2\pi RC}$$

$$= \frac{1}{2\pi \times 20 \times 10^3 \times 1200 \times 10^{-12}}$$

$$= 6631.44 \text{ Hz}$$

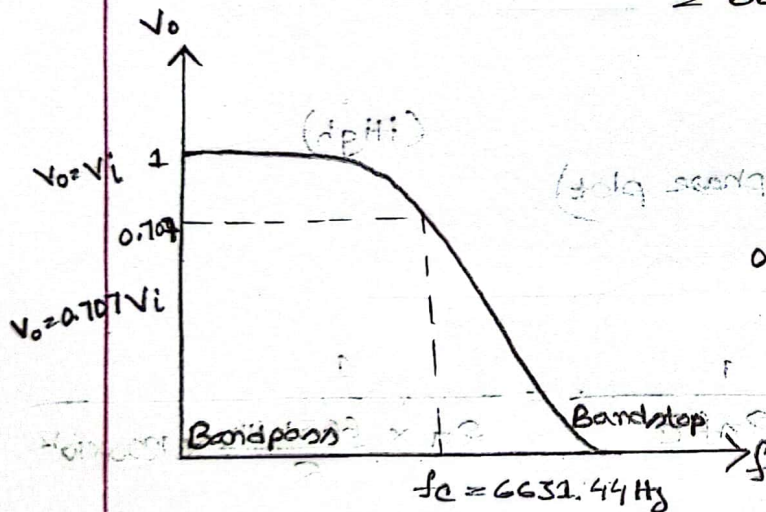


Fig: Normalized plot for low-pass filter

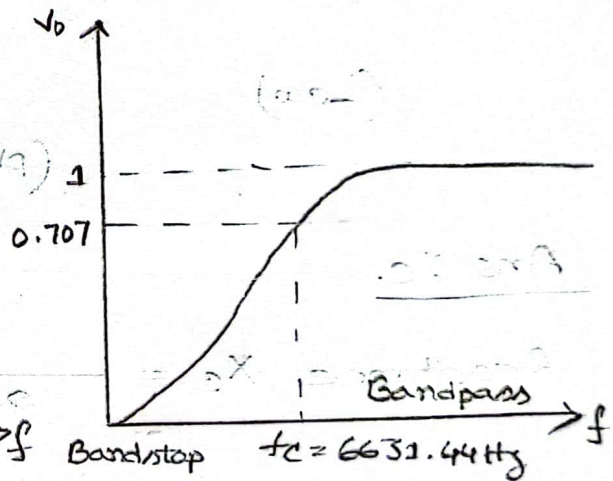
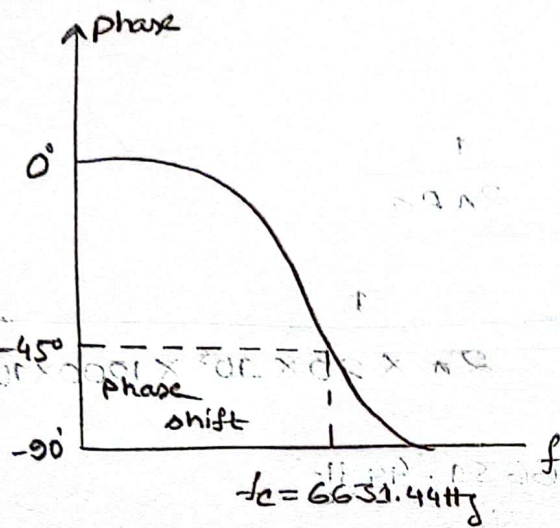


Fig: Normalized plot for High-pass filter

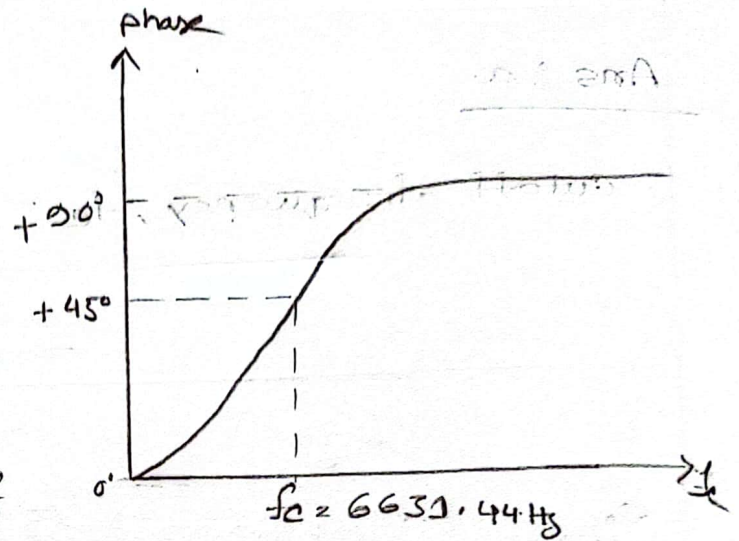
P.T.O

Ans: b.



(Low)

(phase plot)



(High)

Ans: c.

$$\text{Reactance, } X_c = \frac{1}{2\pi f c} = \frac{1}{2\pi \times \frac{6631.44}{2} \times 1200 \times 10^{-12}}$$

$$= 40000 \Omega \approx 40 \text{ k}\Omega$$

$$A_v = \frac{R}{\sqrt{R^2 + X_c^2}} = \frac{20 \times 10^3}{\sqrt{(20 \times 10^3)^2 + (40000)^2}}$$

$$= 0.447$$

$$\therefore \text{phase, } \theta = \tan^{-1} \frac{X_c}{R} = \tan^{-1} \frac{40000}{20 \times 10^3} = 63.43^\circ$$

$$\therefore A_v = 0.447 \angle 63.43^\circ$$

$$f = \frac{1}{2} f_c$$

$$= \frac{1}{2} \times 6631.44$$

$$= 3315.72 \text{ Hz}$$

Band Pass Filter :

A filter that passes a certain band of frequencies to pass through and attenuates others,

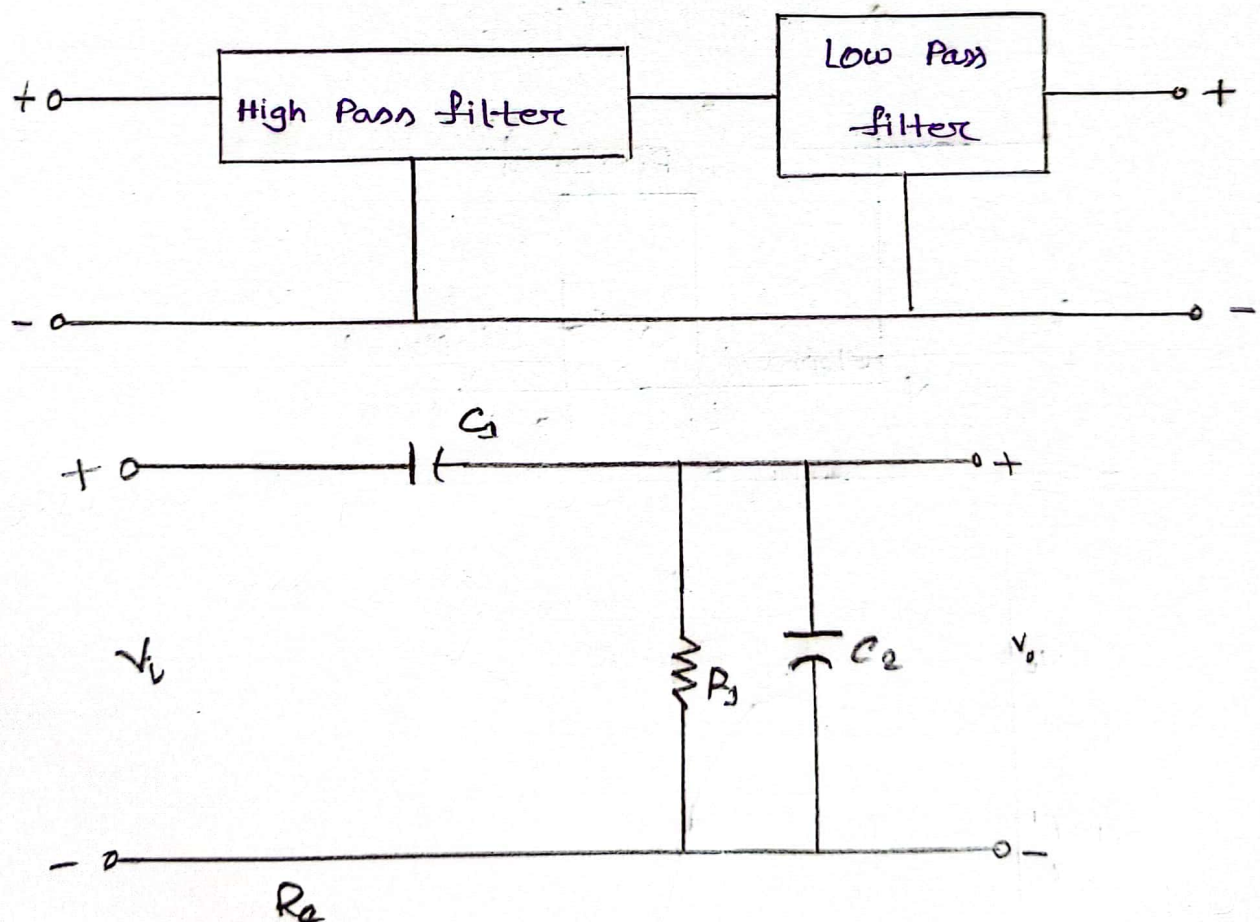


fig: n Bandpass filter using high pass and low pass filter

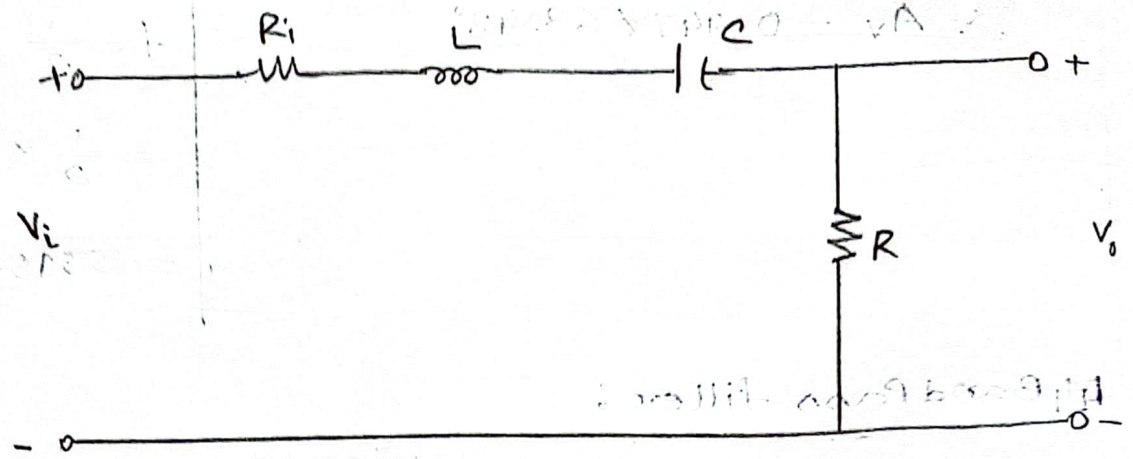
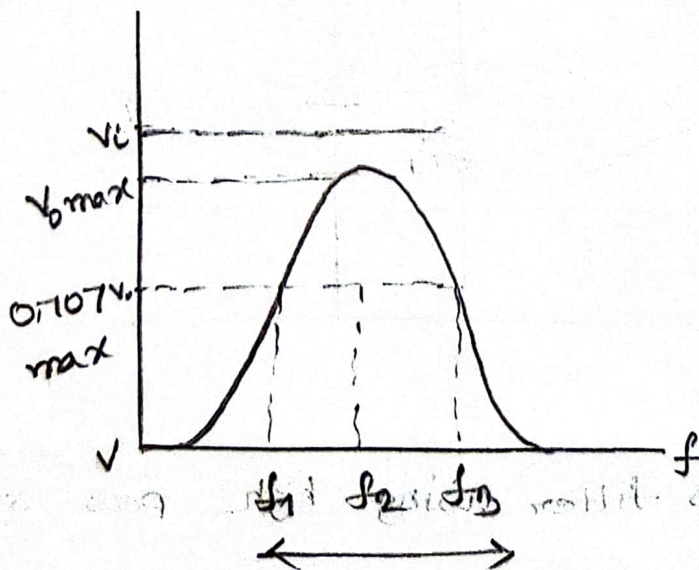
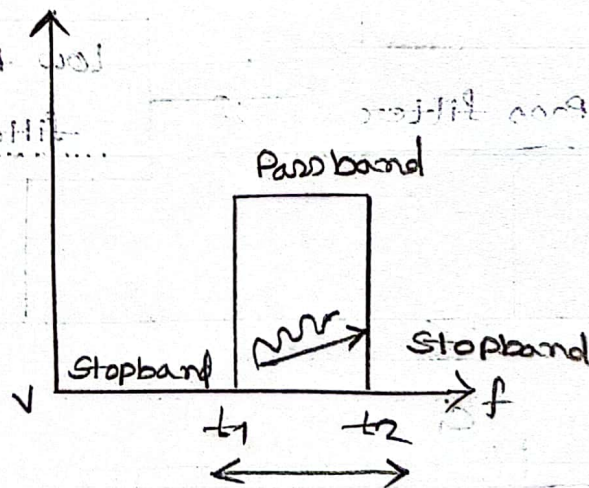


Fig: RL-C Band Pass Filter (Series resonant circuit)



BCW - Bandwidth BandPass

* Frequency when output voltage,

$$\text{is max, } f_s = \frac{1}{2\pi\sqrt{LC}}$$

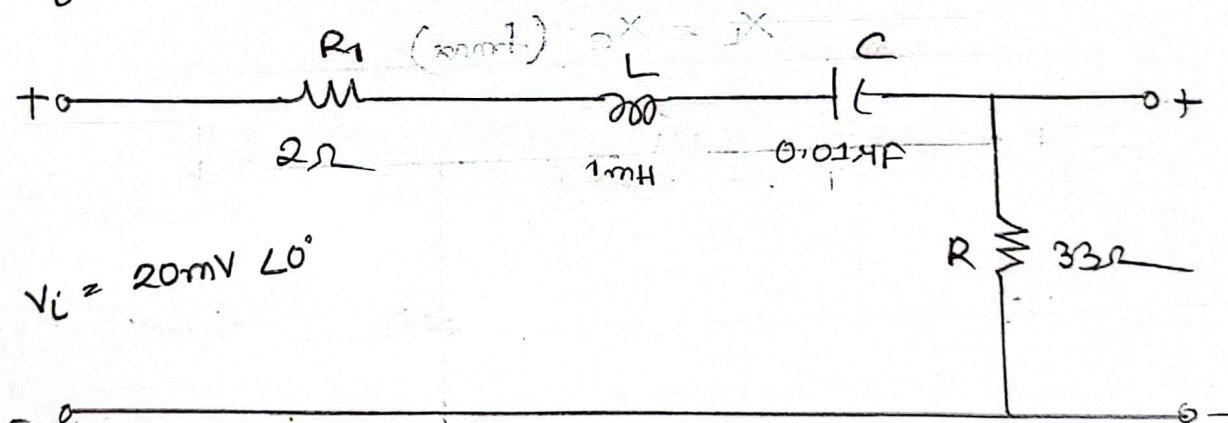
$$X_L = 2\pi f L$$

* Quality factor, $Q_s = \frac{X_L}{R + R_1}$

$$X_C = \frac{1}{2\pi f C}$$

* Bandwidth, $BW = \frac{f_s}{Q_s}$

Q * Example: 22.7 Determine the frequency response for the voltage v_o for the series circuit.



Given, $L = 1 \text{ mH} = 1 \times 10^{-3} \text{ H}$

$C = 0.01 \text{ uF} = 0.01 \times 10^{-6} \text{ F}$

We know,

$$f_s = \frac{1}{2\pi\sqrt{LC}} = \frac{1}{2\pi\sqrt{(1 \times 10^{-3} \times 0.01 \times 10^{-6})}}$$

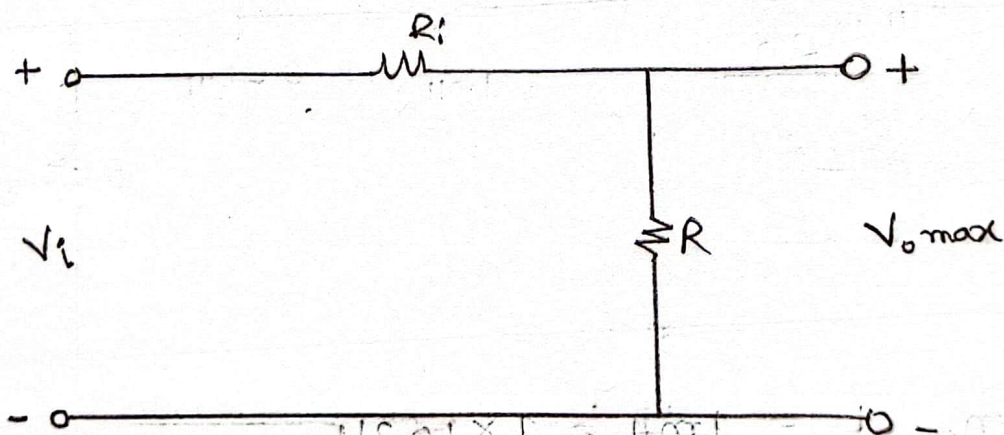
$$= 50,329.21 \text{ Hz}$$

$$Q_s = \frac{X_L}{R + R_i} = \frac{2\pi \times 50,329.21 \times 1 \times 10^{-3}}{2 + 33} = 0.04$$

$$BW = \frac{f_s}{Q_s} = \frac{50,329.21}{0.04} = 5.57 \text{ kHz}$$

Series resonant circuit,

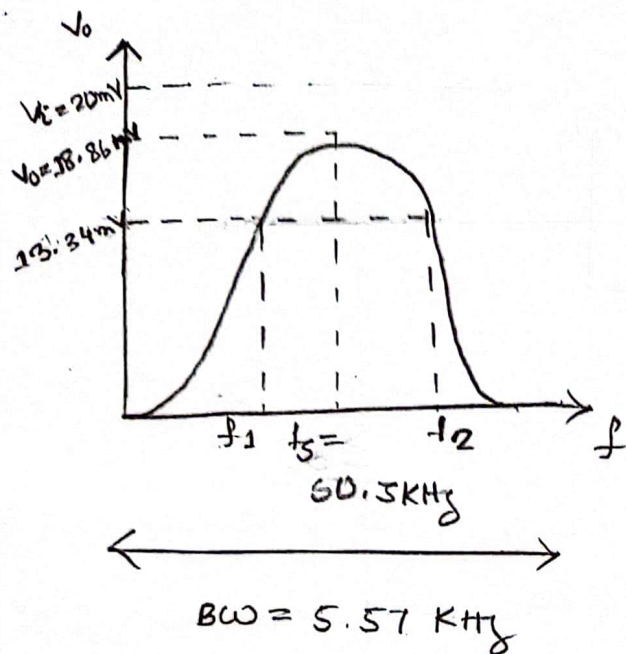
$$X_L = X_C \text{ (at max)}$$



$$V_o \text{ max} = \frac{R}{R_i + R} \times V_i$$

$$= \frac{33}{2 + 33} \times (20 \times 10^{-3})$$

$$\approx 18.86 \text{ mV}$$



Now $\rightarrow 0.707 V_{o \text{ max}} = 0.707 \times 18.86$

$= 13.34 \text{ mV}$

Bandstop Filter :

A bandstop filter passes frequencies outside a frequency band and blocks or attenuates frequencies inside the frequency band.

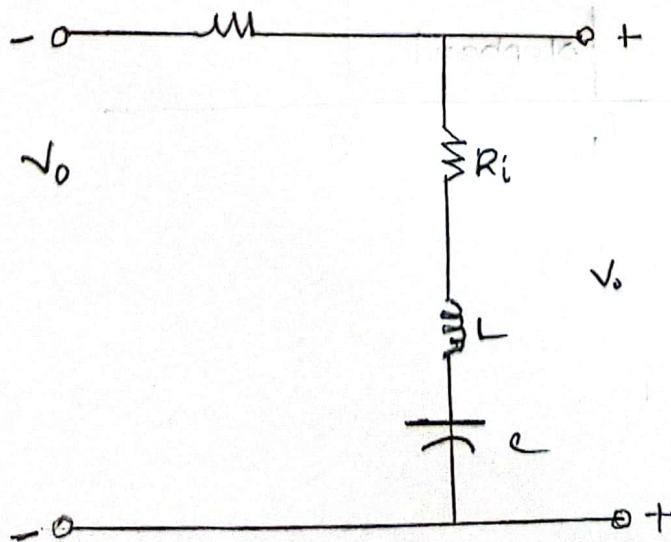


Fig 1. Series resonant circuit of Bandstop filter

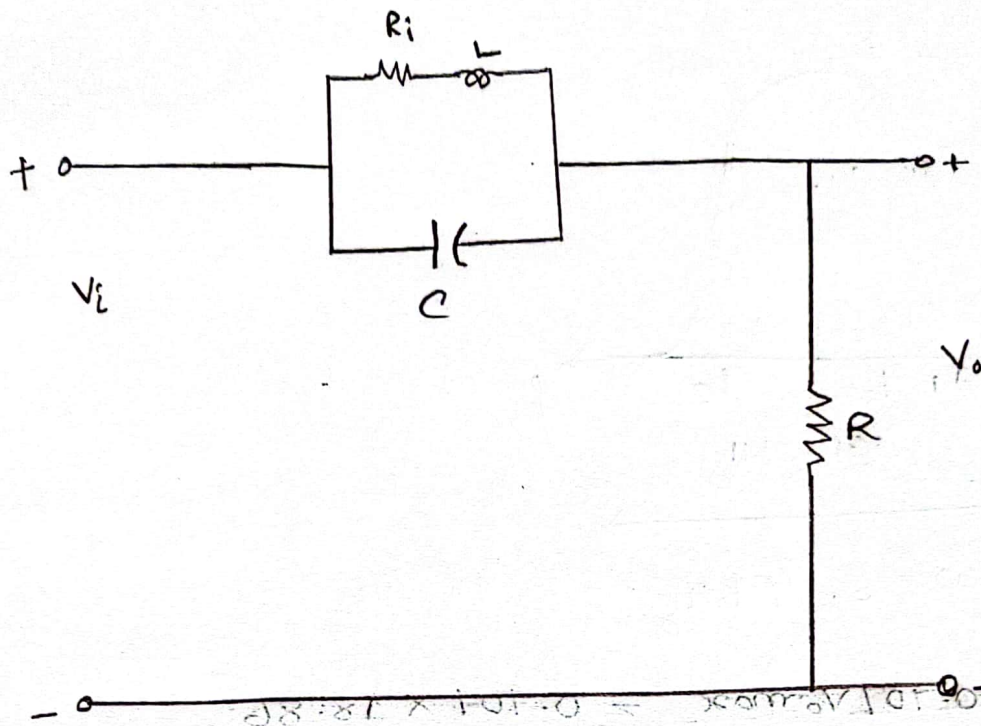
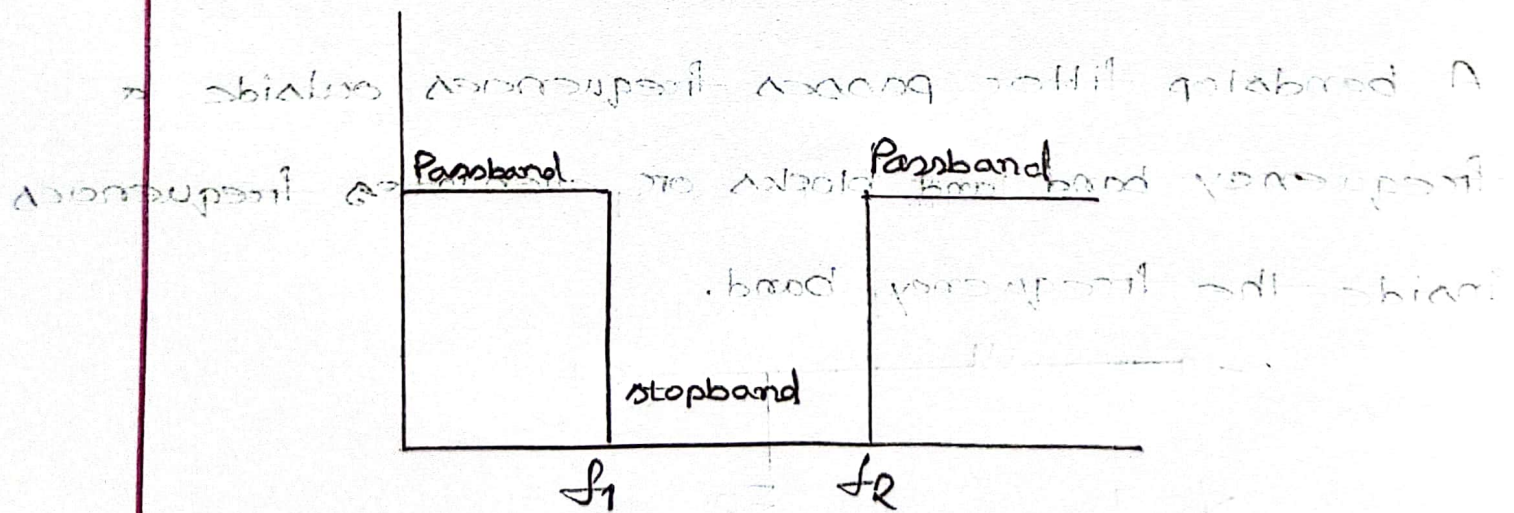


Fig- 02. Parallel resonant circuit of bandstop filter.



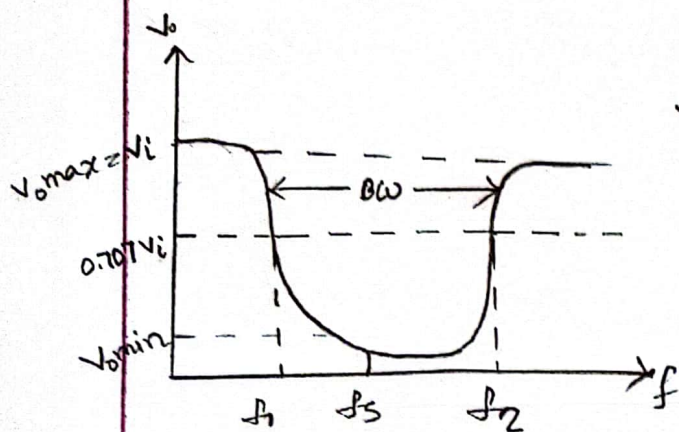


Fig: 03. Frequency response for series circuit

$$\square V_{o \min} = \frac{R_i}{R + R_i} \times V_i$$

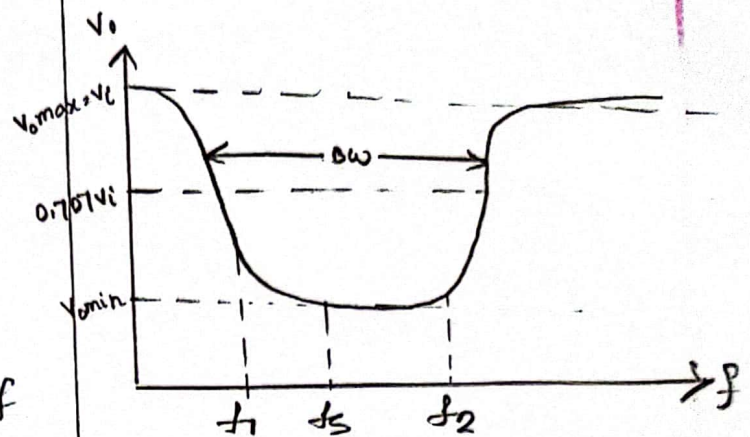


Fig: 04. Frequency response for parallel circuit

$$\square V_{o \min} = \frac{R}{R + Z_{TP}} \times V_i$$

$Z_{TP} = \text{impedance}$